SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

Code of Conduct:

I __T Bhanu Teja(192472259)___ *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Bhanu Teja

QUESTIONS: 05 **Total Marks: 50**

Annexure A

Data Interpretation **Duration: 1hr**

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

1. Reward Trend and Learning Performance

Given

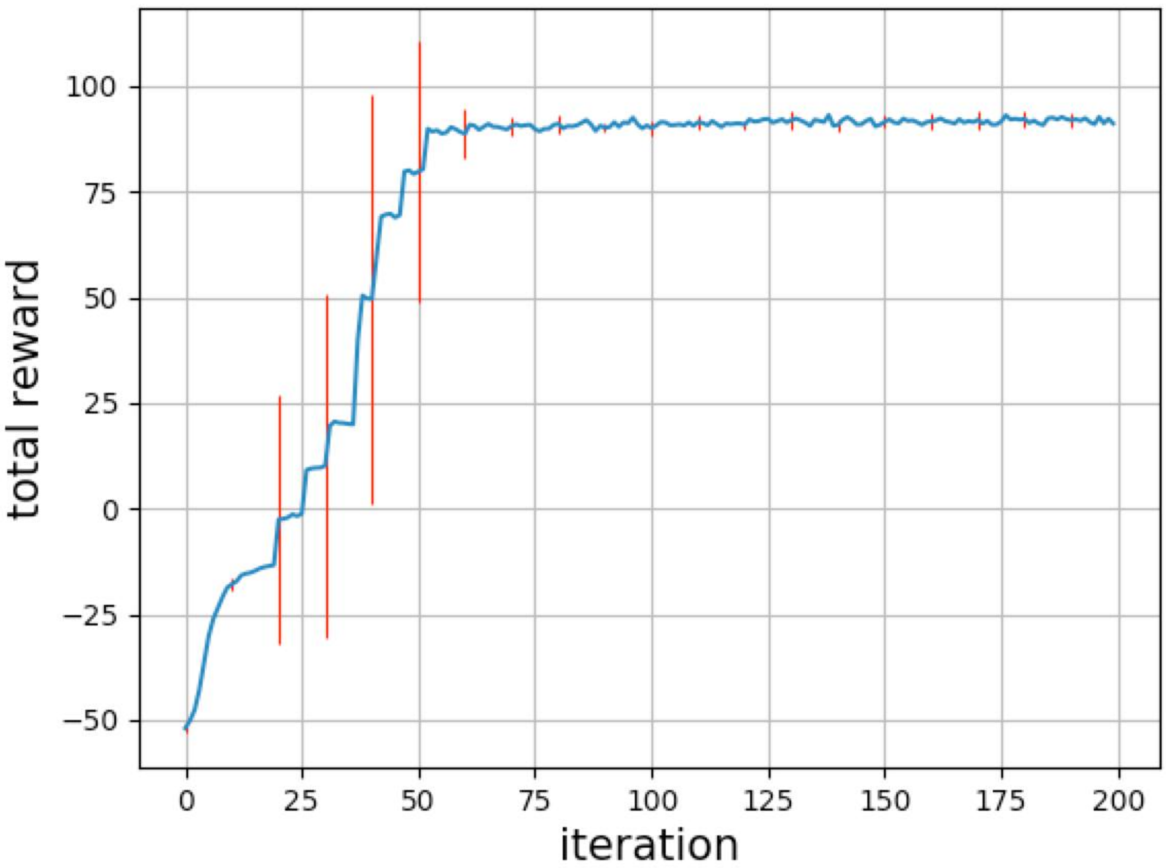
The reward obtained by the RL agent increases gradually from **10 to 45** over multiple episodes.

Answer

In Reinforcement Learning, the **reward** represents the feedback received by an agent after performing an action. The given reward trend shows that the agent's reward increases from **10 initially to 45 after training**. This upward trend indicates that the agent is gradually improving its decision-making ability.

During the initial episodes, the agent has limited knowledge about the environment. Therefore, it may select inappropriate actions and receive lower rewards. As training continues, the agent learns from its previous experiences and starts selecting actions that produce better outcomes.

The gradual increase in reward also indicates that the **policy is improving**. A policy determines which action the agent should take for a particular state. When the policy is updated using successful experiences, the agent becomes more likely to select rewarding actions and avoid poor actions.



The increase from 10 to 45 represents an improvement of:
Thus, the reward has increased by **35 points**, showing significant learning progress.
However, an increasing reward alone does not guarantee that the agent has learned the optimal policy. If the reward eventually becomes stable around a high value, it indicates **convergence**. If it continues increasing, additional training may still improve the policy.

Conclusion

The gradual increase in reward from **10 to 45** demonstrates that the RL agent is successfully learning from experience and improving its policy. The trend indicates positive learning performance, although additional episodes are required to confirm whether the agent has reached an optimal and stable policy.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is

0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

(5 Marks)

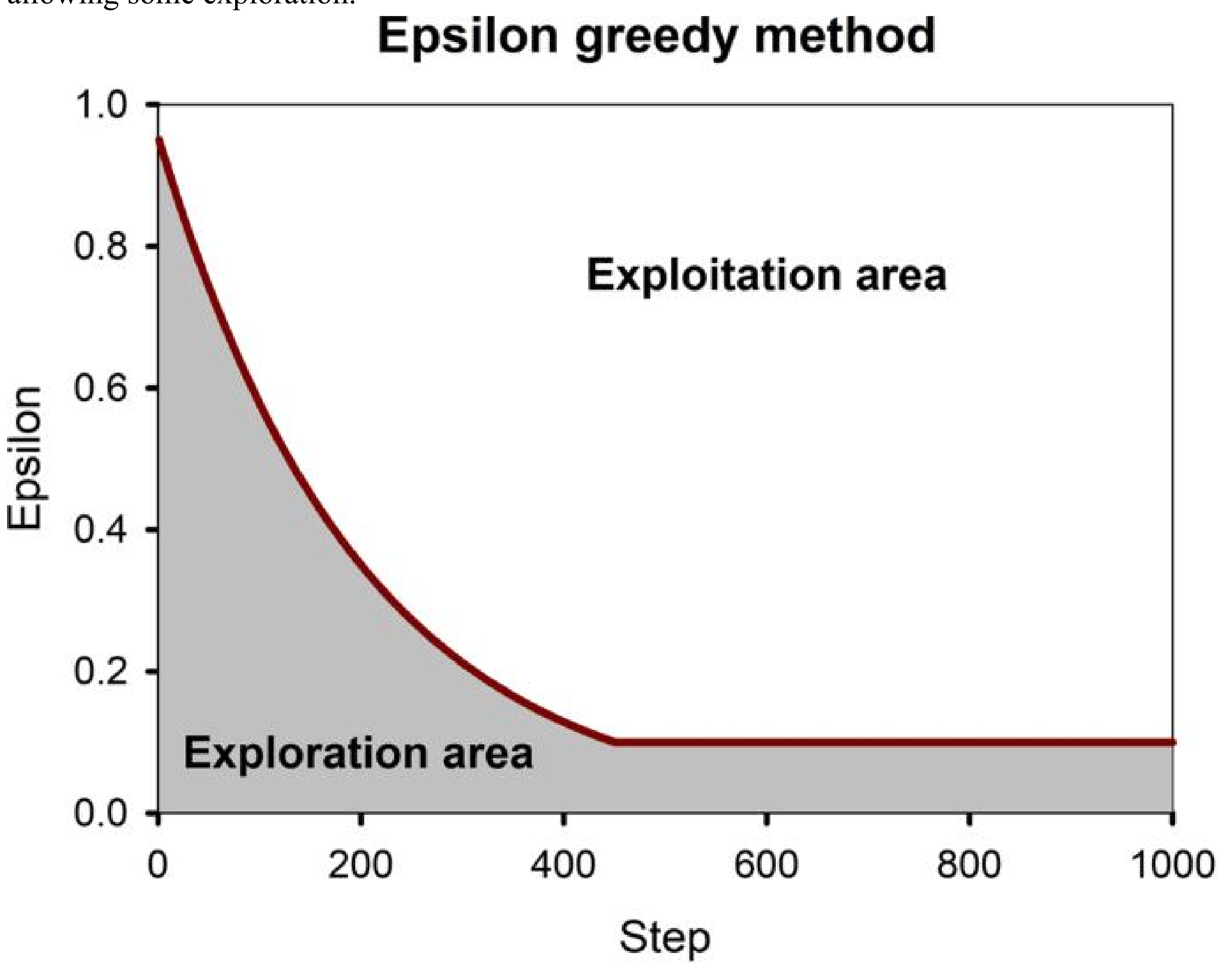
Given

Epsilon (ε)	**Performance**
0.1	Highest cumulative reward
0.3	Medium cumulative reward
0.5	Lowest cumulative reward

Answer

The **exploration–exploitation trade-off** is an important concept in Reinforcement Learning. **Exploration** means trying new or less-known actions to discover potentially better rewards, while **exploitation** means selecting the best-known action based on previous experience.

In the **ε-greedy strategy**, epsilon determines the probability of exploration. With **ε = 0.1**, the agent explores approximately 10% of the time and exploits the best-known action approximately 90% of the time. Therefore, it mainly uses its learned knowledge while still allowing some exploration.



With **ε = 0.3**, the agent explores 30% of the time and exploits 70% of the time. With **ε = 0.5**, exploration and exploitation are equally balanced. This causes the agent to select random actions more frequently.

The given results show that **ε = 0.1 produces the highest cumulative reward**, while **ε = 0.5 produces the lowest reward**. This suggests that, in this particular environment, excessive exploration reduces performance because the agent frequently chooses actions that are not yet known to be rewarding.

However, an epsilon value that is too small can also be problematic because the agent may fail to discover better actions. Therefore, a suitable balance is required.

Conclusion

Based on the observed results, **ε = 0.1 provides the best performance** because it favors exploitation while maintaining limited exploration. The experiment demonstrates that the optimal epsilon value depends on the environment and the amount of knowledge already learned by the agent.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance

difference between these algorithms and determine which algorithm converges faster based on the observed results.

(5 Marks)

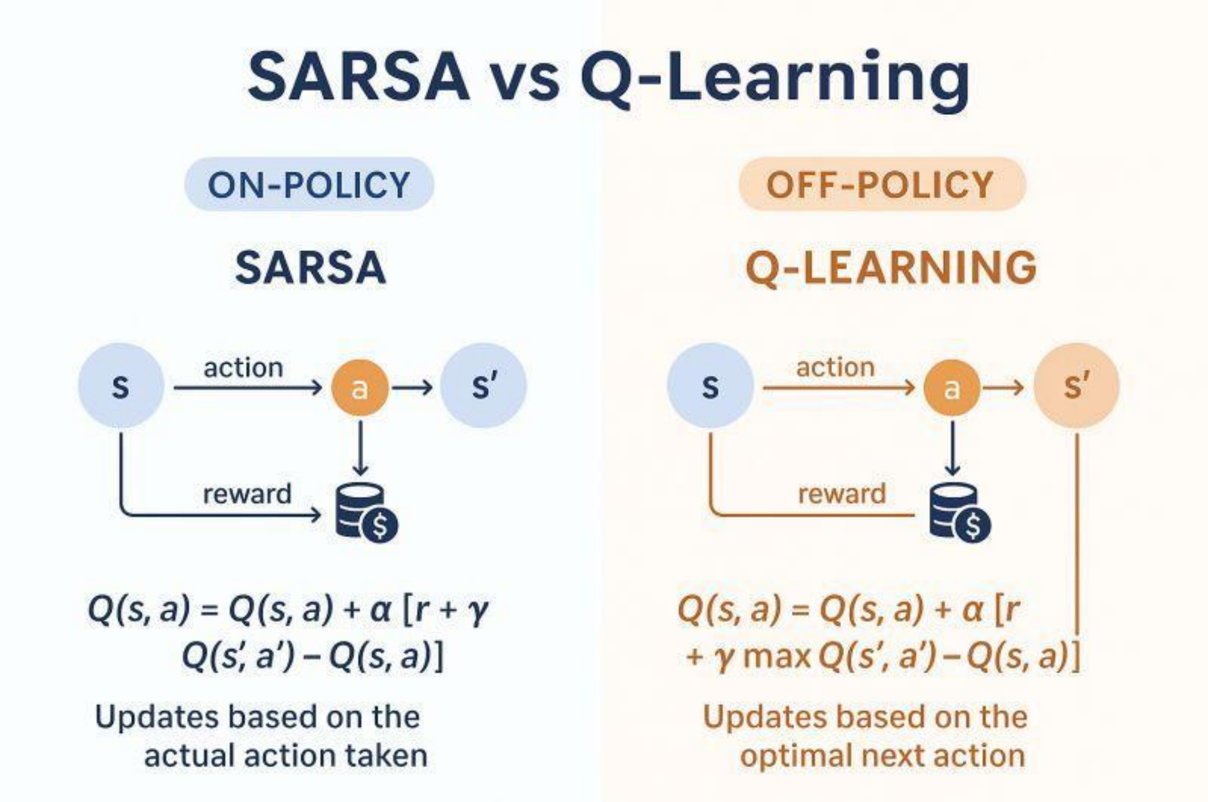
Given

Q-Learning and SARSA are evaluated at **100, 200, and 300 episodes**, and Q-Learning consistently obtains slightly higher rewards.

Answer

Q-Learning and **SARSA** are both **Temporal Difference (TD) Reinforcement Learning algorithms**, but they use different approaches for updating their value estimates.

Q-Learning is an **off-policy** algorithm. It learns the optimal action-value function by considering the action with the maximum future Q-value, regardless of which action the agent actually follows. Therefore, it tends to learn an optimal policy more aggressively.



SARSA is an **on-policy** algorithm. Its update considers the actual next action selected by the current policy. Therefore, SARSA learns the value of the policy that the agent is actually following, including its exploration behavior.

The observed results show that Q-Learning consistently produces slightly higher rewards at **100, 200, and 300 episodes**. This indicates that Q-Learning is performing better in the given environment.

Episodes	Q-Learning	SARSA	Observation
100	Higher	Lower	Q-Learning performs better
200	Higher	Lower	Difference continues
300	Higher	Lower	Q-Learning maintains advantage

Since Q-Learning maintains higher rewards throughout the experiment, it can be considered to have **better learning performance** in this particular case. Its convergence may also be faster if its reward becomes stable earlier than SARSA.

However, the exact convergence speed cannot be determined only from the statement that Q-Learning has higher rewards. A reward-versus-episode graph would provide stronger evidence.

Conclusion

Q-Learning performs slightly better than SARSA based on the observed rewards. Its off-policy nature allows it to directly learn toward the best estimated action. Therefore, **Q-Learning appears more effective in this experiment**, although the environment and hyperparameters can affect the final comparison.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance.

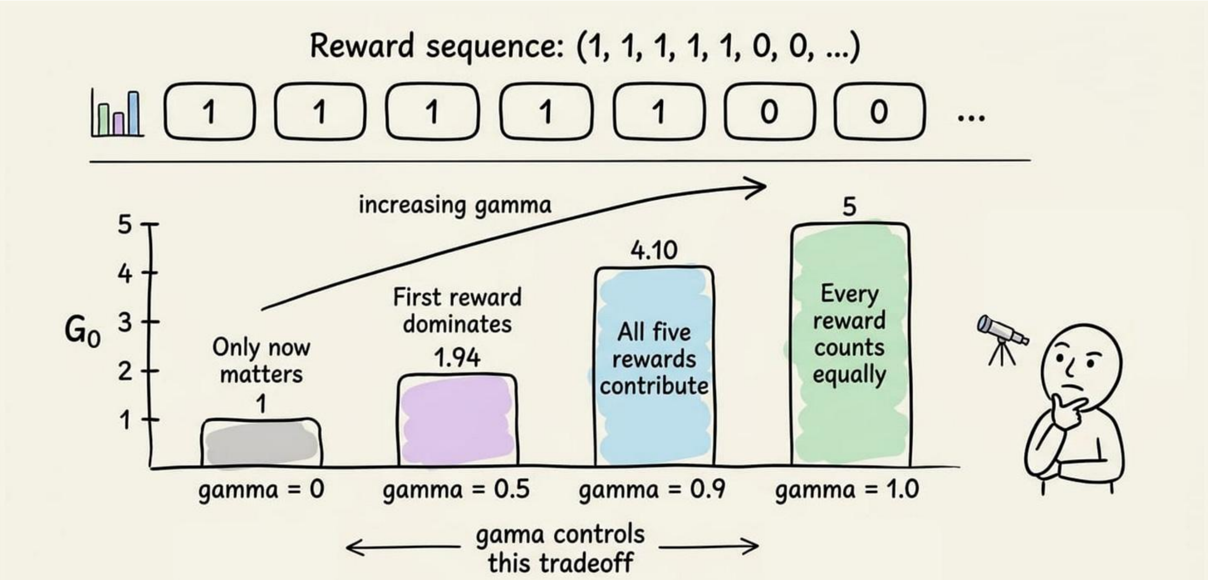
(5 Marks)

Given

The agent is trained using:

- $\gamma = 0.2$
- $\gamma = 0.5$
- $\gamma = 0.9$

The average reward increases as γ increases.
Answer
The discount factor (γ) determines how much importance an RL agent gives to future rewards compared with immediate rewards. Its value normally ranges from 0 to 1.
When $\gamma = 0.2$, the agent gives greater importance to immediate rewards and gives relatively little importance to future rewards. This makes the agent behave more short-sightedly.



When $\gamma = 0.5$, the agent considers both immediate and future rewards with moderate importance. It provides a balance between short-term and long-term decision-making.
When $\gamma = 0.9$, the agent gives significant importance to future rewards. Therefore, it may accept a smaller immediate reward if doing so can produce greater rewards later.
The experiment shows that the average reward increases as γ increases. This indicates that the problem benefits from long-term planning. The agent performs better when it considers the future consequences of its actions.

Discount Factor	Behavior	Observed Performance
0.2	Focuses mainly on immediate rewards	Lower
0.5	Balanced short/long-term view	Medium
0.9	Strong focus on future rewards	Highest

Therefore, among the tested values, $\gamma = 0.9$ provides the best performance.
Conclusion
The increasing reward with higher discount factors shows that long-term decision-making is important in this environment. Hence, $\gamma = 0.9$ is the most suitable value among the tested options. However, an excessively high discount factor may not always be suitable for every RL problem.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

(5 Marks)

Given

- Success rate at **50 episodes = 45%**
- Success rate at **200 episodes = 92%**

Answer

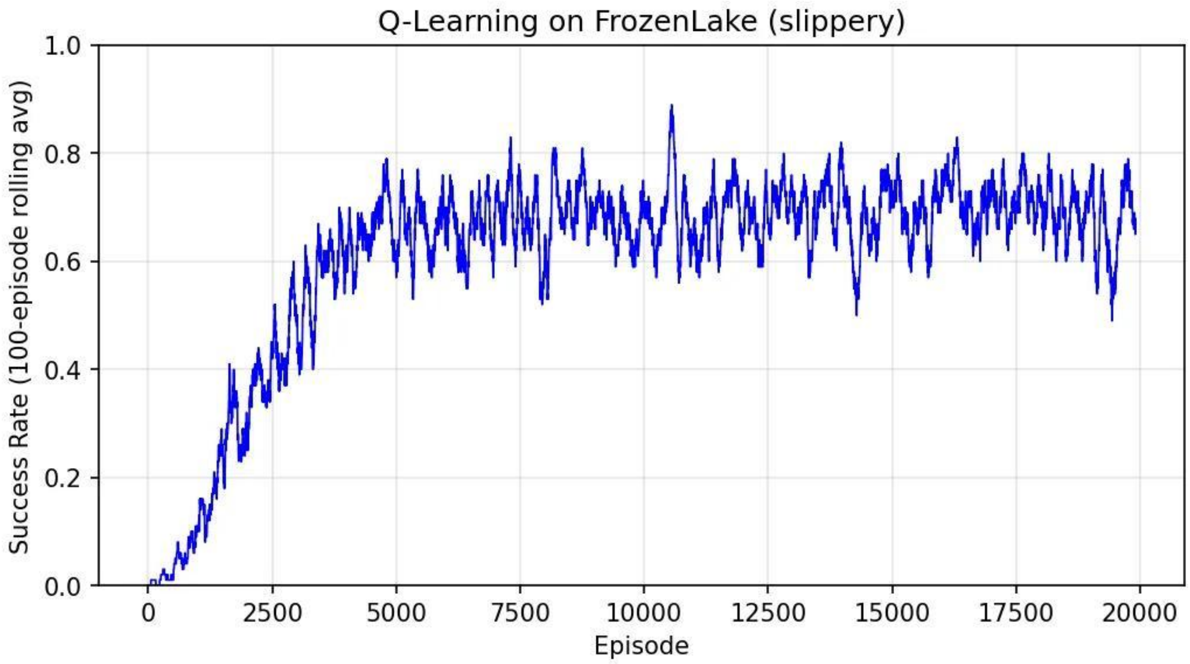
The **success rate** represents the percentage of tasks or episodes in which the RL agent successfully achieves its objective. In this experiment, the success rate increases from **45% at 50 episodes to 92% at 200 episodes**.

The improvement in success rate is:

Thus, there is a **47 percentage-point improvement** in the agent's success rate.

This significant improvement indicates that the agent is learning from its interactions with the environment. During the early 50 episodes, the agent may make many incorrect decisions because it has limited knowledge. After additional training, it learns which actions are more effective and begins following a better policy.

The success rate of **92%** indicates strong performance. It means the agent successfully completes the task in approximately 92 out of every 100 episodes under the evaluated conditions.



However, a 92% success rate alone does **not prove that the policy is optimal**. To determine whether the agent has reached an optimal policy, we should observe whether the success rate becomes stable over additional episodes. We can also compare the result with the maximum achievable success rate.

Conclusion

The increase from **45% to 92%** demonstrates substantial learning and policy improvement. The agent has learned a highly effective policy, but further training and convergence analysis are required to confidently conclude that it has reached the **optimal policy**.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand